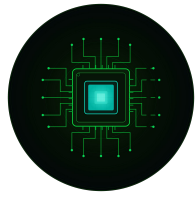

Is Moore's Law Dead?

What Lies Beyond Transistor Scaling

The Evolution of Electronics

Transistor History & The Birth of Modern Computing



1906

Vacuum Tubes

First amplifying devices. Large, hot, unreliable — but enabled radios and early computers like ENIAC.

1947

Transistor Invented

Bell Labs. Shockley, Bardeen & Brattain. Smaller, cooler, faster. The revolution begins.

1958

Integrated Circuit

Jack Kilby (TI) & Robert Noyce (Fairchild). Multiple transistors on one chip — the IC age.

1971

Intel 4004

First commercial microprocessor. 2,300 transistors at 10 μm . Fits in your palm. Changes everything.



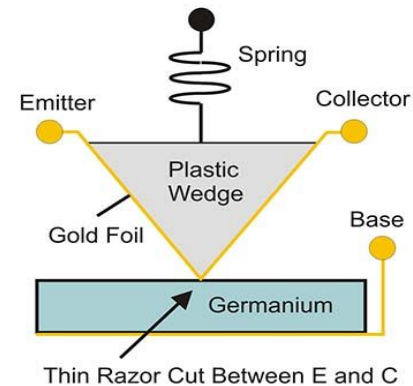
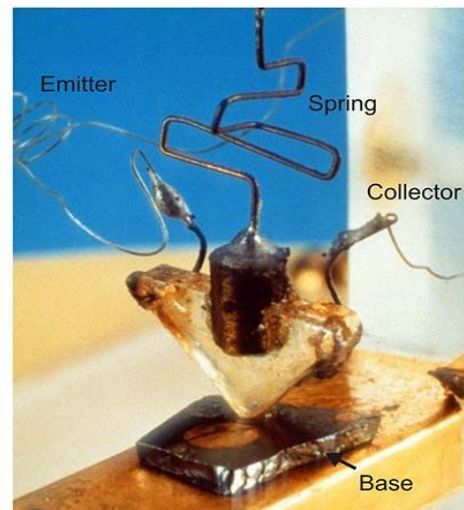
From room-sized machines to billions of transistors on a fingernail — miniaturization drove the entire digital age.



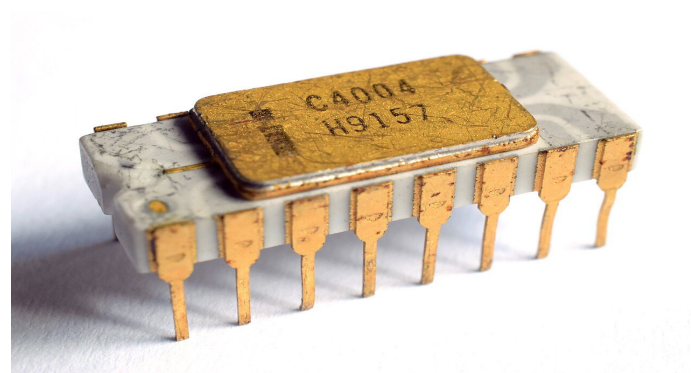
Vacuum tubes for 1st generation computers.



The first integrated circuit by Jack Kilby

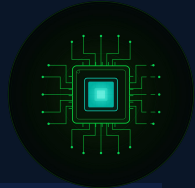


The first transistor



Intel 4004

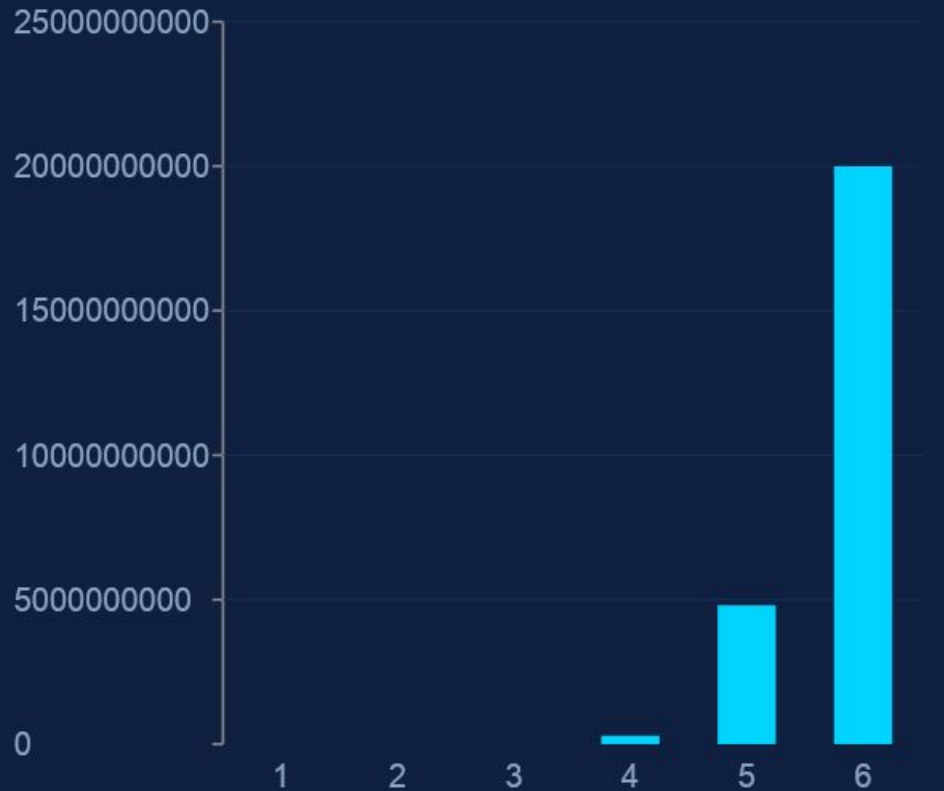
Moore's Law & Exponential Scaling



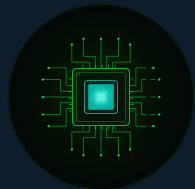
Gordon Moore's Prediction (1965)

- Transistor count doubles roughly every 2 years
- Originally observed, later became a self-fulfilling industry roadmap
- Each generation: 2× transistors, same cost
- Smaller = faster switching, lower voltage, less power per operation

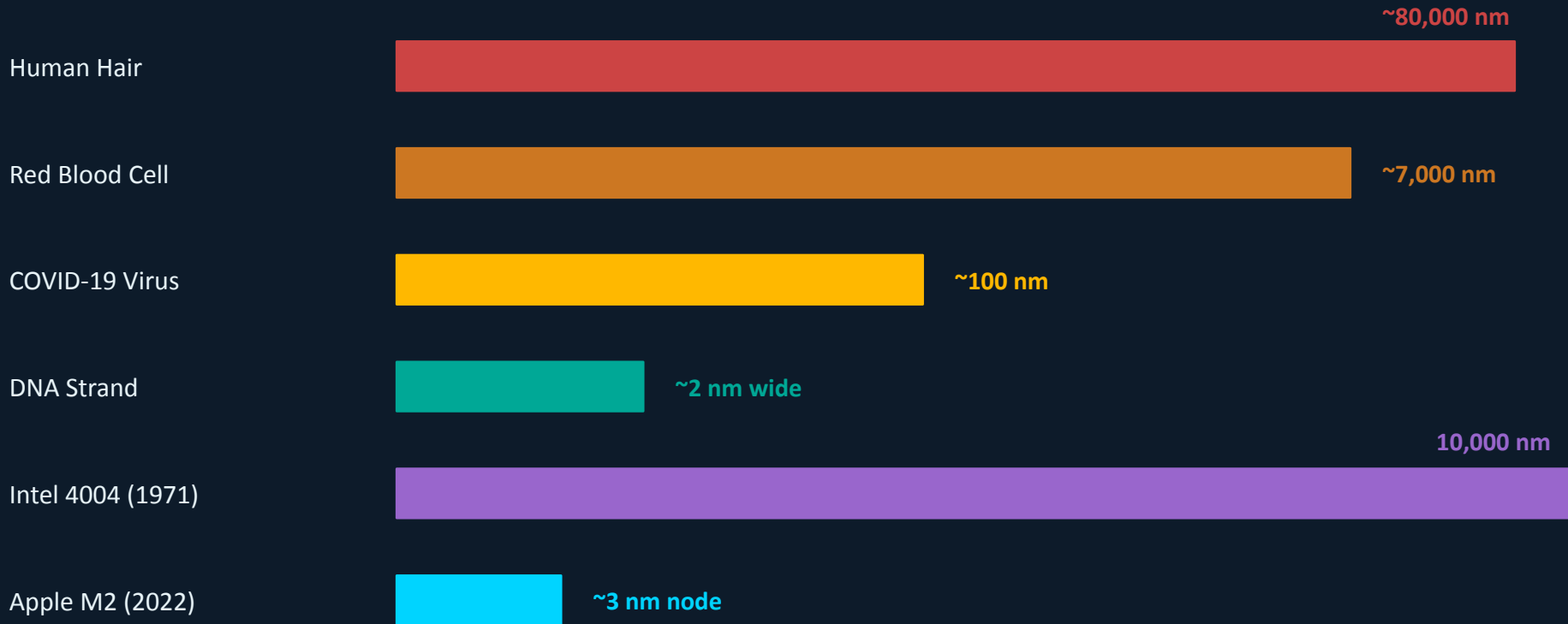
1971 → 2024: Transistor count grew ~1 Trillion ×



How Small Is a Modern Transistor?

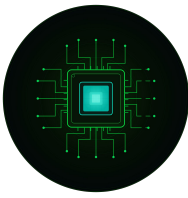
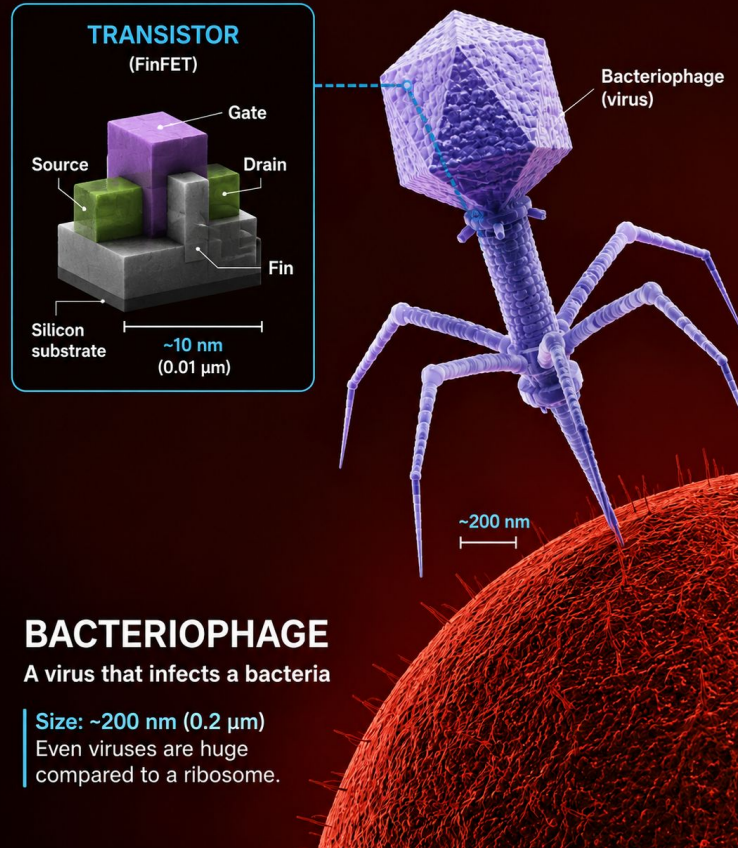


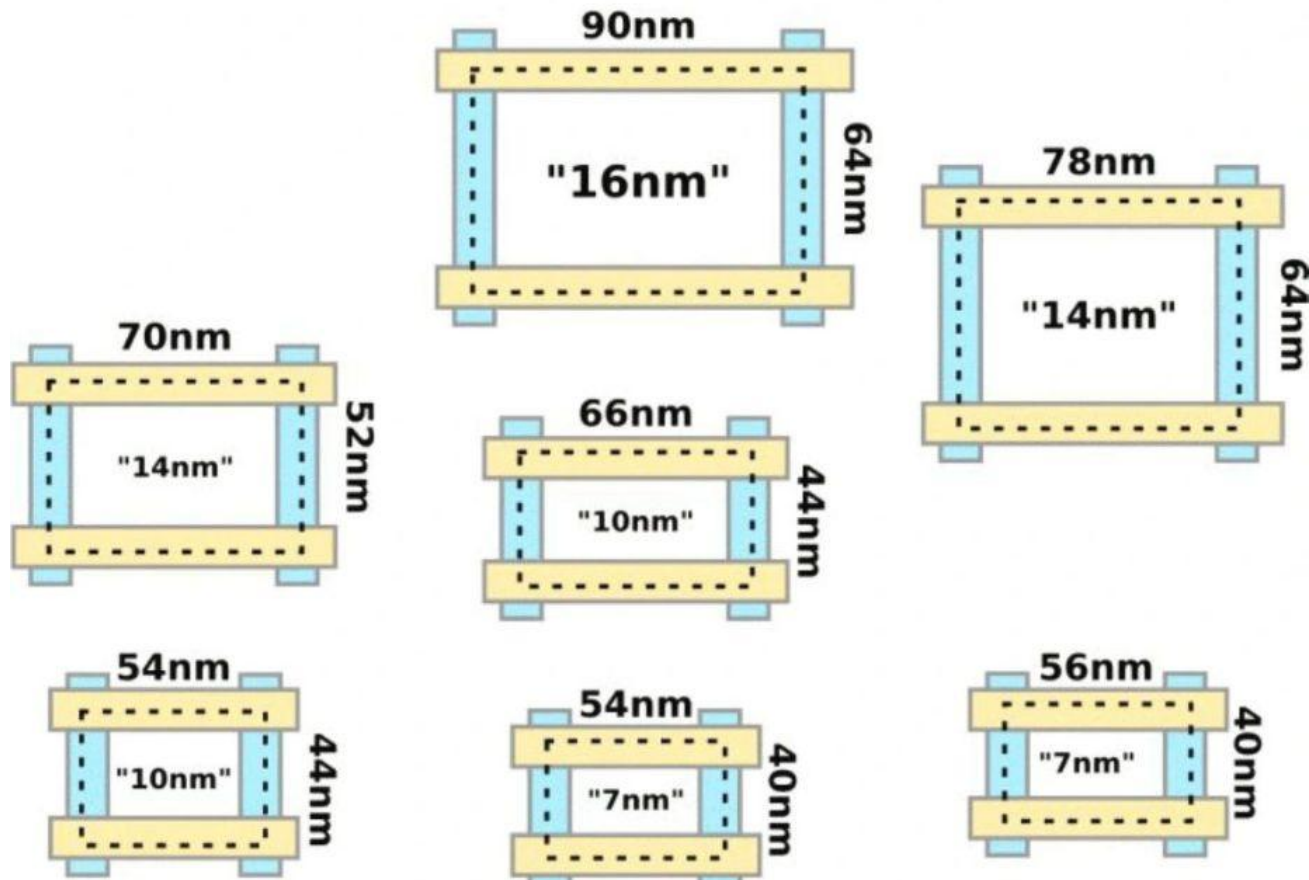
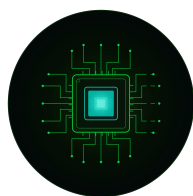
MIND-BLOWN MOMENT :) · Atomic-Scale Engineering



A 3nm transistor gate is ~15 silicon atoms wide. We are literally building with individual atomic layers.

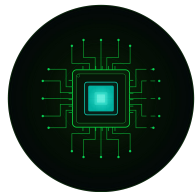
This is how **tiny** a translator
really is—**Smaller** than a virus.





Why Dennard Scaling Broke Down

Heat · Leakage · Quantum Effects · Lithography Challenges



Dennard's Rule (1974)

Shrink transistors → keep power density constant. Smaller = faster AND cooler. This worked until ~2005.



Heat & Power Density

Voltage couldn't scale down as fast as transistors. Power/area exploded — modern CPUs hit 100W/cm², hotter than a nuclear reactor core per unit area.



Leakage Current

Ultra-thin gate oxide (a few atoms) lets electrons quantum-tunnel through even when OFF. Wasted power grows exponentially with shrinking.



Quantum Effects

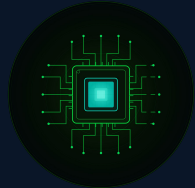
At sub-10nm, quantum tunneling, threshold voltage variation, and statistical dopant fluctuations cause random errors.



Lithography Wall

UV light wavelength can't print features smaller than ~100nm. Required radical new approaches (ArF immersion, then EUV).

Transistor Architecture Evolution



Planar → FinFET → GAAFET / Nanosheet

Planar (pre-2011)

130nm – 22nm

Gate contact: 1 side

✓ Advantages

- + Simple 2D structure
- + Mature process
- + Easy manufacturing

✗ Limitations

- Gate on 1 side only
- Severe leakage below 28nm
- Can't scale further

FinFET (2011–now)

22nm – 3nm

Gate contact: 3 sides

✓ Advantages

- + 3D fin: gate wraps 3 sides
- + 10× better leakage control
- + Enabled scaling to 3nm

✗ Limitations

- Fin width limited to ~5nm
- Difficult below 3nm
- Parasitic capacitance rises

GAAFET / Nanosheet

2nm and below

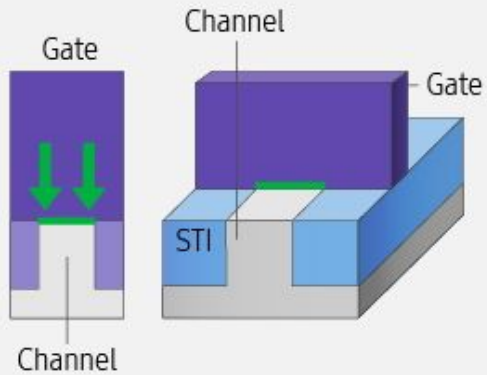
Gate contact: All 4 sides

✓ Advantages

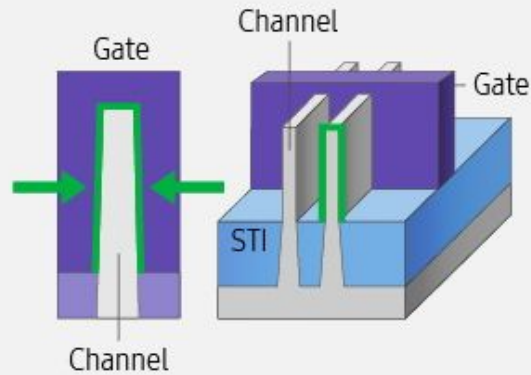
- + Gate surrounds all 4 sides
- + Maximum electrostatic control
- + Tunable sheet width

✗ Limitations

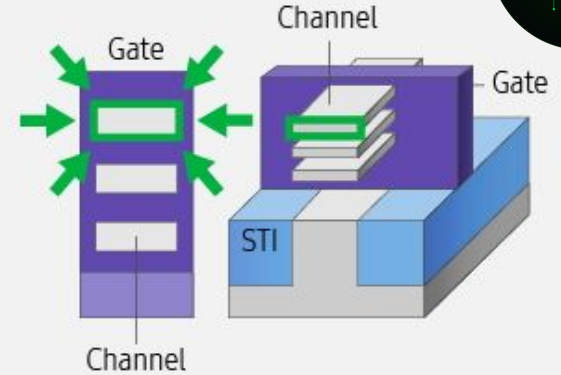
- Extremely complex process
- New materials needed
- Yield challenges



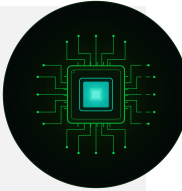
Planar FET
1 Gate on channel



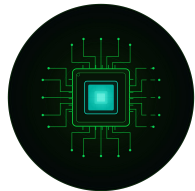
FinFET
3 Gates on channel



Gate-All-Around
4 Gates on channel



EUV Lithography & Why ASML Is Irreplaceable



WOW MOMENT · The Most Precise Machines Ever Built

How EUV Works

- 1 A powerful CO₂ laser fires 50,000×/sec at a tin droplet 30 μm wide
- 2 This vaporizes the tin and emits EUV light at 13.5nm wavelength
- 3 The light bounces off 6 ultra-precise mirrors (the flattest surfaces ever made)
- 4 Each mirror is so smooth that if scaled to Earth's size, bumps would be <1mm tall
- 5 The patterned light exposes a wafer — printing features <8nm wide
- 6 The entire machine operates in near-perfect vacuum (EUV absorbed by air)

\$380M+

Cost of one EUV machine

13.5 nm

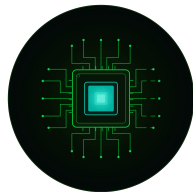
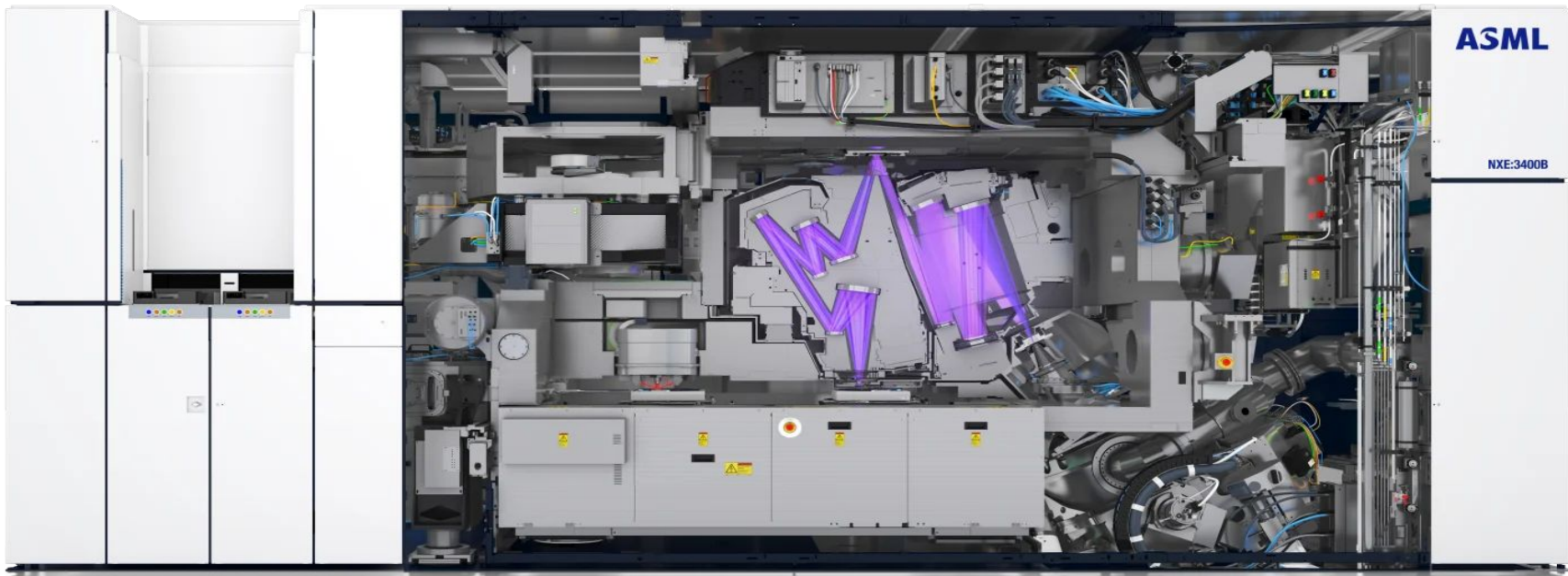
EUV wavelength (vs 193nm for ArF)

1 company

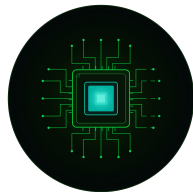
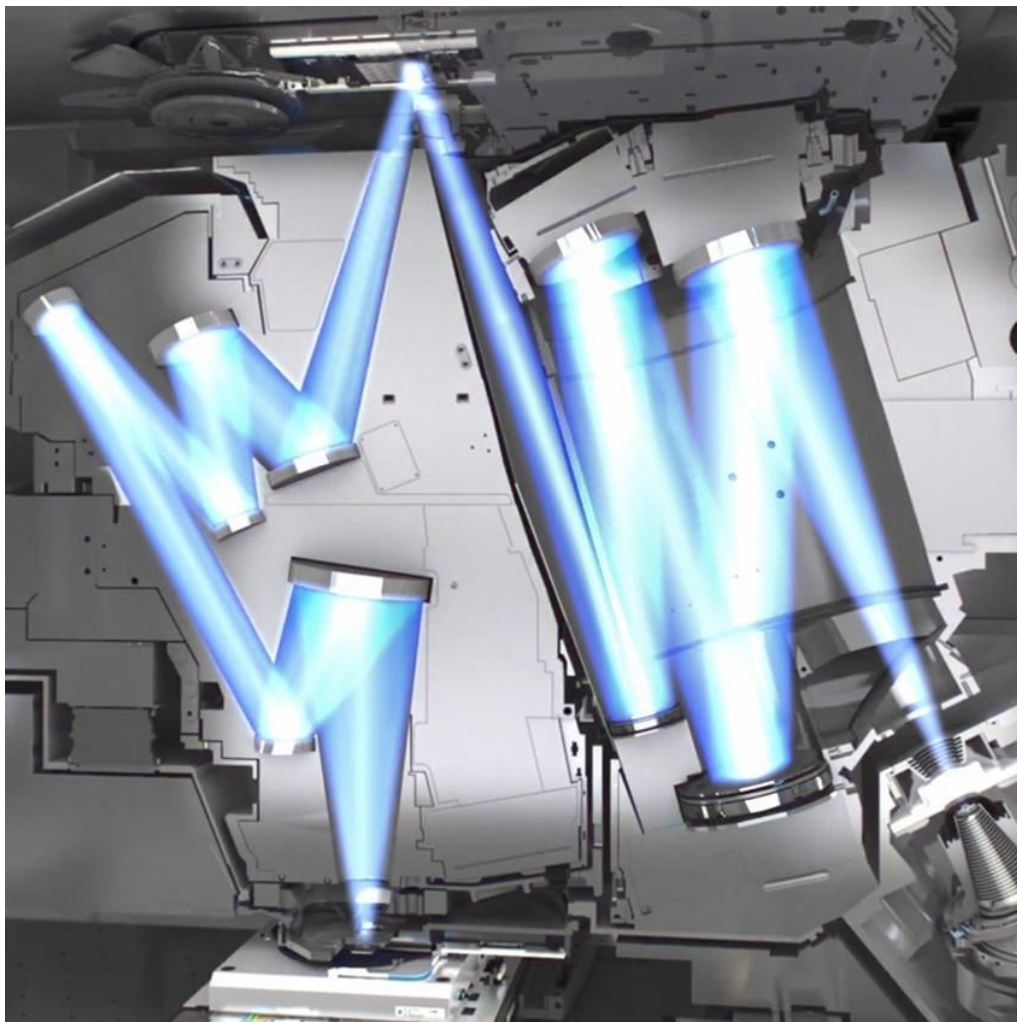
ASML — sole global supplier

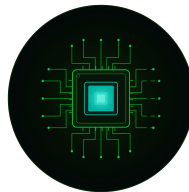
<0.1 nm

Mirror surface roughness

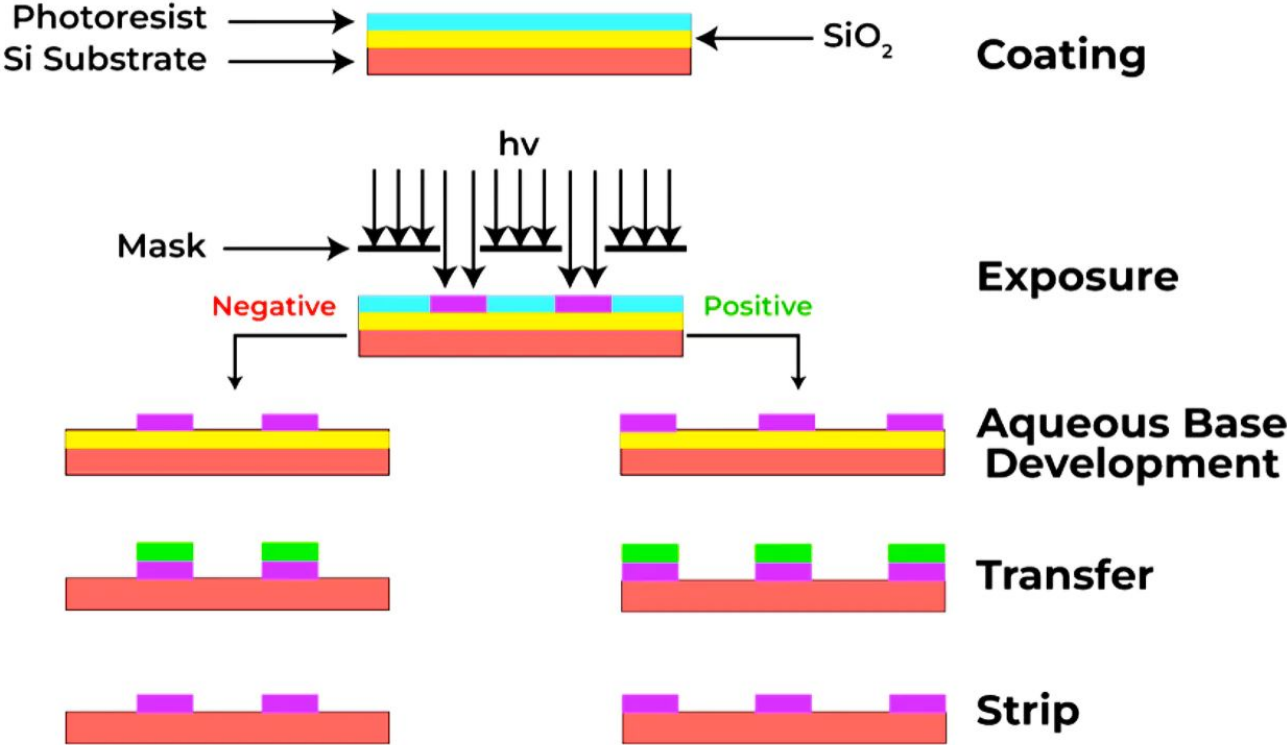


Cost: ~\$200- \$400 million each.



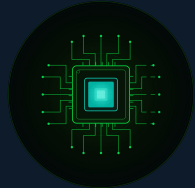


Photolithographic Process

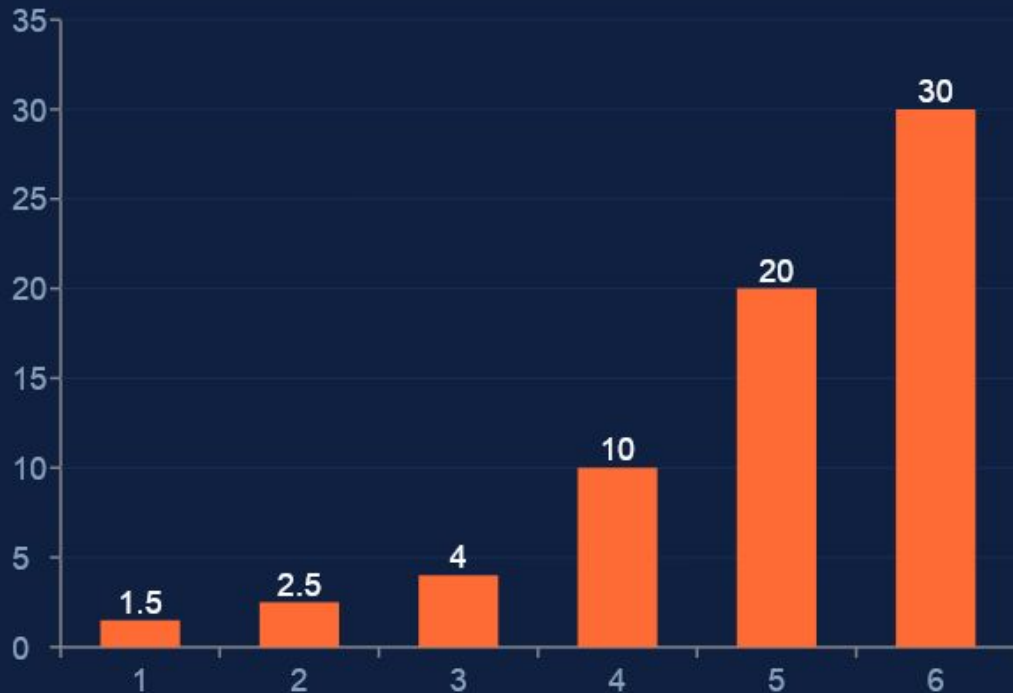


Photolithography Process

Why Fabs Are Becoming Impossibly Expensive



The Most Expensive Buildings in Human History



\$30B+

Cost of TSMC 2nm fab (est.)

~1,000

Process steps per wafer

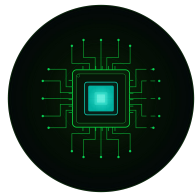
< 3 companies

Can build leading-edge fabs



Rock's Law: fab cost doubles every 4 years. Moore's Law squares off against an economic wall.

Why Architecture Innovation Matters More Now



The Interconnect Bottleneck & Rise of Specialized Hardware

WOW MOMENT: Interconnect delay now exceeds transistor switching delay

Moving data between transistors takes longer than the transistors themselves switching. The wire is the bottleneck, not the switch.

Domain-Specific Architectures

GPUs for parallel math, TPUs for matrix multiply, NPU's for neural nets. 10-100× efficiency vs general-purpose CPUs for their target workload.



Memory Bandwidth as Priority

HBM stacked memory brings data closer to compute. Bandwidth, not clock speed, limits AI training performance.



Dataflow & Systolic Arrays

Reuse data in-place, avoid moving it to/from memory. Google TPU's systolic array computes matrix ops with near-zero memory traffic.

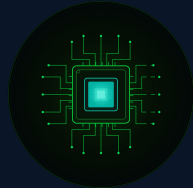


In-Memory Computing

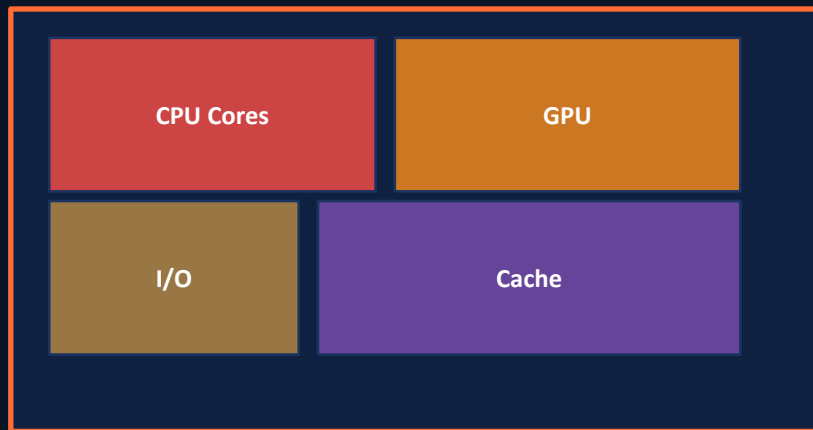
Compute where data lives — resistive RAM arrays performing MAC operations. Eliminates the von Neumann bottleneck entirely.

The Chiplet Revolution

Modular Silicon — Mixing Nodes, Vendors, and Functions

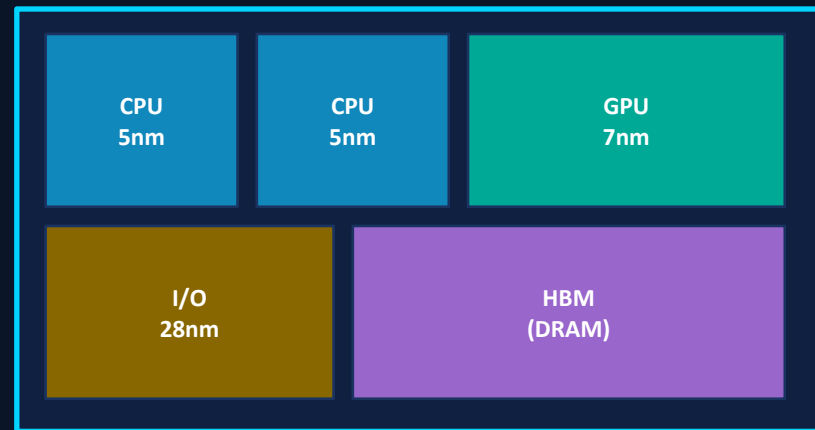


Monolithic Die (Old)



One die, one node. A defect anywhere = full scrap.

Chiplet Design (New)



Mix best nodes per function. Better yield. AMD, Intel, Apple all use chiplets.

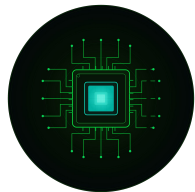
✓ Higher Yield

✓ Mix Process Nodes

✓ Modular Reuse

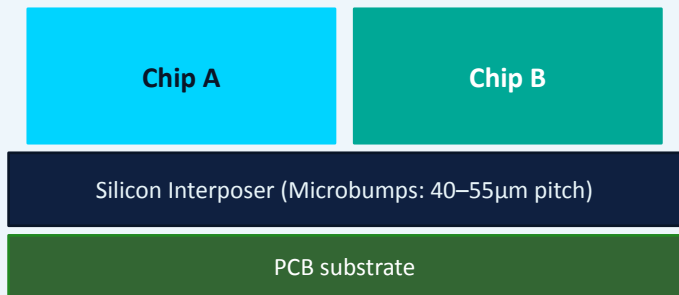
✓ Faster Time-to-Market

2.5D Packaging & 3D Stacking



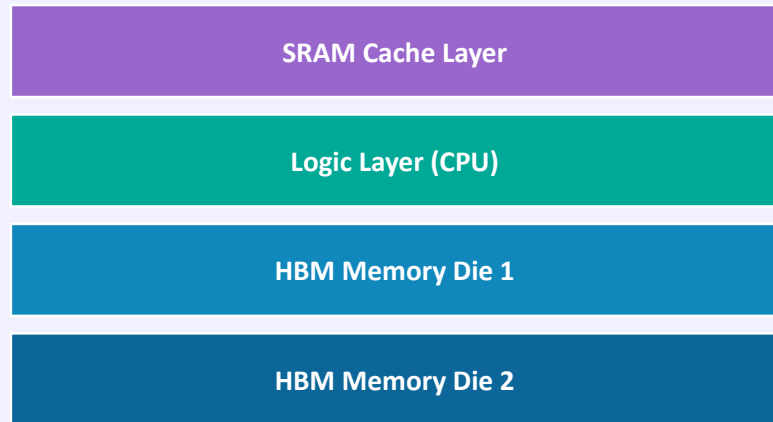
WOW MOMENT · Packaging Is Now as Important as Transistor Technology

2.5D — Silicon Interposer



NVIDIA H100 uses CoWoS: 4 HBM3 stacks on a silicon interposer → 3.35 TB/s memory bandwidth.

3D — Monolithic Stacking



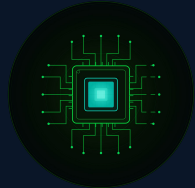
Through-Silicon Vias (TSVs) connect layers with millions of vertical wires — sub-micron diameter.



Advanced packaging is now a primary competitive differentiator — as critical as the transistor node itself.

What Lies Beyond Transistor Scaling?

Emerging Technologies on the Horizon



Early R&D

Quantum Computing

Qubits exploit superposition and entanglement. Exponential speedup for specific problems (cryptography, simulation). IBM, Google, IonQ racing ahead. Not a CPU replacement — a co-processor.

Research

Neuromorphic Chips

Brain-inspired architectures. Intel Loihi 2, IBM TrueNorth. Spiking neural networks — event-driven, ultra-low power. 1000× more efficient than GPUs for sparse inference.

Emerging

Photonic Computing

Use photons instead of electrons. Light travels at c , generates zero heat from resistance. Huge bandwidth. Challenges: on-chip light sources, modulators, and memory.

Lab Stage

Carbon Nanotubes

CNT transistors switch faster than silicon at same size. IBM demonstrated 10,000-transistor CNT chips. Manufacturing yield remains the key challenge.

Research

2D Materials (MoS₂)

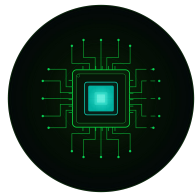
Molybdenum disulfide is 1 atom thick. Perfect electrostatics. Could extend scaling past silicon's physical limits. MIT, Stanford active research.

Concept

DNA & Molecular Computing

Data storage in DNA: 1 gram = 215 PB. Computation via strand displacement. Not general-purpose but massively parallel for specific tasks.

Key Takeaways



01

Moore's Law $\neq$ Dead, But Slowing — Transistor counts still grow, but frequency and power efficiency gains have plateaued since ~2005. The era of 'free lunch' scaling is over.

02

Physical Limits Are Real — Quantum tunneling, heat density, and lithography barriers are fundamental physics — not engineering problems to engineer away.

03

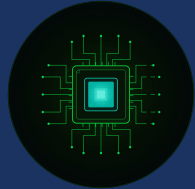
Architecture Is the New Frontier — Specialization (GPUs, TPUs, NPUs), chiplets, and 3D packaging deliver more performance-per-watt than further shrinking alone.

04

Packaging Is Now a First-Class Citizen — 2.5D and 3D stacking, silicon interposers, and TSVs are as competitively important as transistor node generation.

05

The Supply Chain Is Geopolitical — ASML's EUV monopoly, TSMC's fab dominance, and rare materials mean semiconductor technology is national security infrastructure.



Thank You

Questions & Discussion

Topics covered: Transistor history · Moore's & Dennard's Laws · FinFET/GAAFET · EUV Lithography · Fab economics · Chiplets · 3D Stacking · Future technologies